

EE 102: Signal Processing and Linear Systems

Problem Set #3: LTI Systems

Name: _____

Submission Date: _____

Problem 1 [Adapted from Lathi 1.1-11] Consider a signal that is a sum of complex exponentials given by

$$x(t) = \sum_{k=m}^n D_k e^{j\omega_k t}$$

where $D_k \in \mathbb{C}$ and $\{\omega_k\}$ are pairwise distinct, that is, $\omega_i \neq \omega_j$ for all $i \neq j$.

(a) [5 points] Show that the time-averaged power of $x(t)$ is

$$P_\infty(x) = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt = \sum_{k=m}^n |D_k|^2.$$

Hint: expand $|x(t)|^2$ and use that $\frac{1}{2T} \int_{-T}^T e^{j(\omega_k - \omega_\ell)t} dt \rightarrow 0$ when $\omega_k \neq \omega_\ell$.

(b) [5 points] Determine $E_\infty(x) = \int_{-\infty}^{\infty} |x(t)|^2 dt$. State clearly whether it is finite or infinite for this $x(t)$ and justify.

(c) [5 points] Let

$$\tilde{x}(t) \triangleq \overline{x(t)} = \sum_{k=m}^n \overline{D_k} e^{-j\omega_k t}.$$

What is the power for this conjugate signal?

(d) [5 points] Is $x(t)$ even, odd, or neither? If it is not necessarily even/odd, state the sufficient conditions on $\{D_k, \omega_k\}$ under which $x(t)$ becomes even or odd. For a challenge, you can attempt to show both necessary and sufficient conditions.

[use more pages if needed]

Problem 2 [Adapted from Vierinen 7.1-1]

You often hear that “noise cancelling” works by playing a signal that is completely out of phase with the noise, thus cancelling its perception to the listener. In this problem, you will be asked to mathematically analyze (and prove) this concept. **You must use complex exponentials to solve this problem.** Consider a noise signal given by

$$x_{\text{noise}}(t) = \sin(\omega t + \phi).$$

A cancellation signal of the *same frequency* is

$$x_{\text{cancel}}(t) = \sin(\omega t + \phi_c).$$

(a) [10 points] Find a value of ϕ_c such that $x_{\text{noise}}(t) + x_{\text{cancel}}(t) \equiv 0$. Assume equal amplitudes.

(b) [15 points] Now suppose

$$x_{\text{noise}}(t) = A \sin(\omega t + \phi), \quad x_{\text{cancel}}(t) = B \sin(\omega t + \phi_c),$$

with $A, B > 0$ not *a priori* equal. Using complex exponentials, determine conditions on B and ϕ_c for *perfect* cancellation for all t . If $B \neq A$ is fixed, is perfect cancellation possible?

(c) [5 points] Propose a simple LTI system that, when driven by $x_{\text{noise}}(t) = \sin(\omega_0 t + \phi)$, produces an output $y(t)$ that cancels the noise when added to $x_{\text{noise}}(t)$. Draw a block diagram of your system.

[use more pages if needed]

Problem 3 [Adapted from Lathi P1.7-4] Consider the following system that models an operational amplifier with guard rails $\pm V_{\text{ref}}$ and a propagation delay $t_p > 0$:

$$y(t + t_p) = \begin{cases} -V_{\text{ref}}, & x(t) > V_{\text{ref}}, \\ V_{\text{ref}}, & x(t) < -V_{\text{ref}}, \\ -x(t), & \text{otherwise.} \end{cases}$$

Equivalently, letting $\text{clip}(v) = \max(-V_{\text{ref}}, \min(v, V_{\text{ref}}))$,

$$y(t + t_p) = -\text{clip}(x(t)).$$

Prove/disprove the following:

- (a) [5 points] Sketch the static input-output curve and mark the slopes.
- (b) [5 points] Is the system causal?
- (c) [5 points] Is it invertible? If not, characterize the largest input set on which it *is* invertible.
- (d) [5 points] Is it linear?
- (e) [5 points] Is it time-invariant? Prove or give a counterexample.
- (f) [5 points] Compute and sketch $y(t)$ for $x(t) = A \cos(\omega t)$ with $A > V_{\text{ref}}$ (i.e., a clipped sinusoid), and indicate the time fraction spent at each rail.

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Problem 4 [Adapted from Lathi 1.7-11]: A challenging problem, but one that will solidify your understanding!

For a certain LTI system, the output $y(t)$ depends on the input $x(t)$ and on the initial states of the system, specified by two values $q_1(0)$ and $q_2(0)$. The following experiments are performed:

$x(t)$	$q_1(0)$	$q_2(0)$	$y(t)$
0	1	-1	$e^{-t}u(t)$
0	2	1	$e^{-t}(3t+2)u(t)$
$u(t)$	-1	-1	$2u(t)$

The input $x(t)$ is an externally applied signal. The values $q_1(0)$ and $q_2(0)$ describe the state in which the system begins.

(a) [6 points] Note that for the first two experiments (the first two rows), the input is zero, so we have $x(t) = 0$. Using linearity, determine the output of the system for

$$(q_1(0), q_2(0)) = (1, 0)$$

Hint: Recall the pre-requisites! Can you express $(1, 0)$ as a linear combination of the initial states in the first two experiments?

(b) [6 points] Determine the output that would result from

$$q_1(0) = q_2(0) = -1$$

when $x(t) = 0$

Use this result together with the third experiment to determine the output produced by the input $x(t) = u(t)$ when

$$q_1(0) = q_2(0) = 0.$$

(c) [8 points] Now let $q_1(0) = q_2(0) = 0$ and apply a pulse signal as the input, that is, $x(t) = u(t) - u(t - 10)$.

Using linearity and time invariance, find $y(t)$.

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